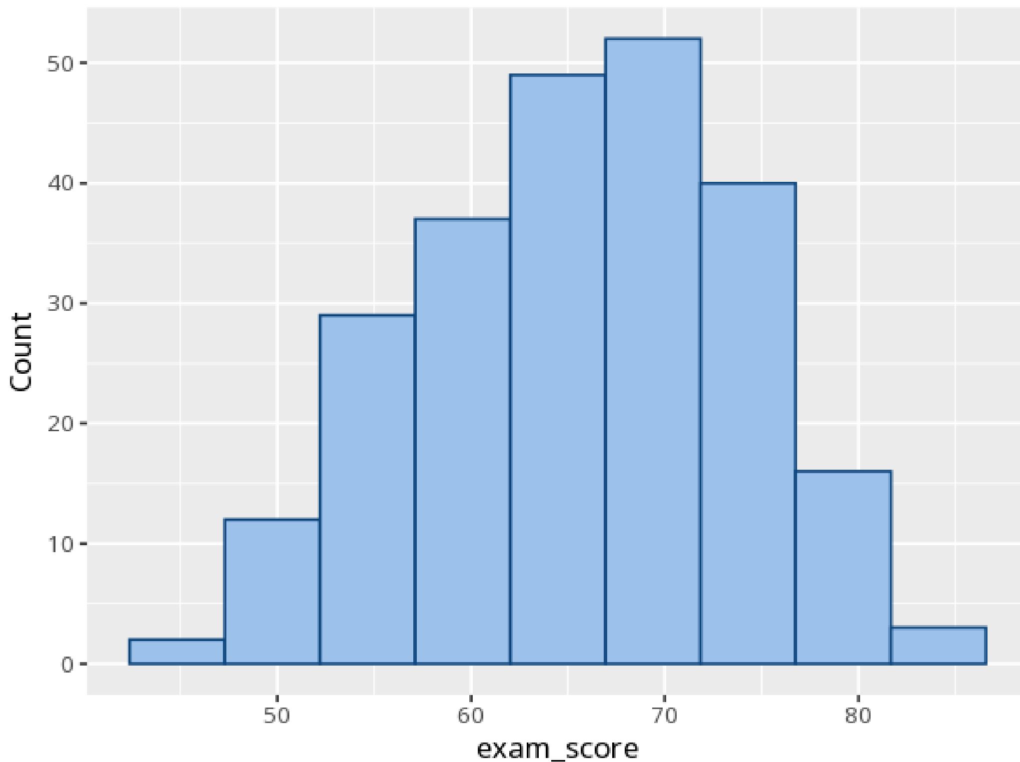
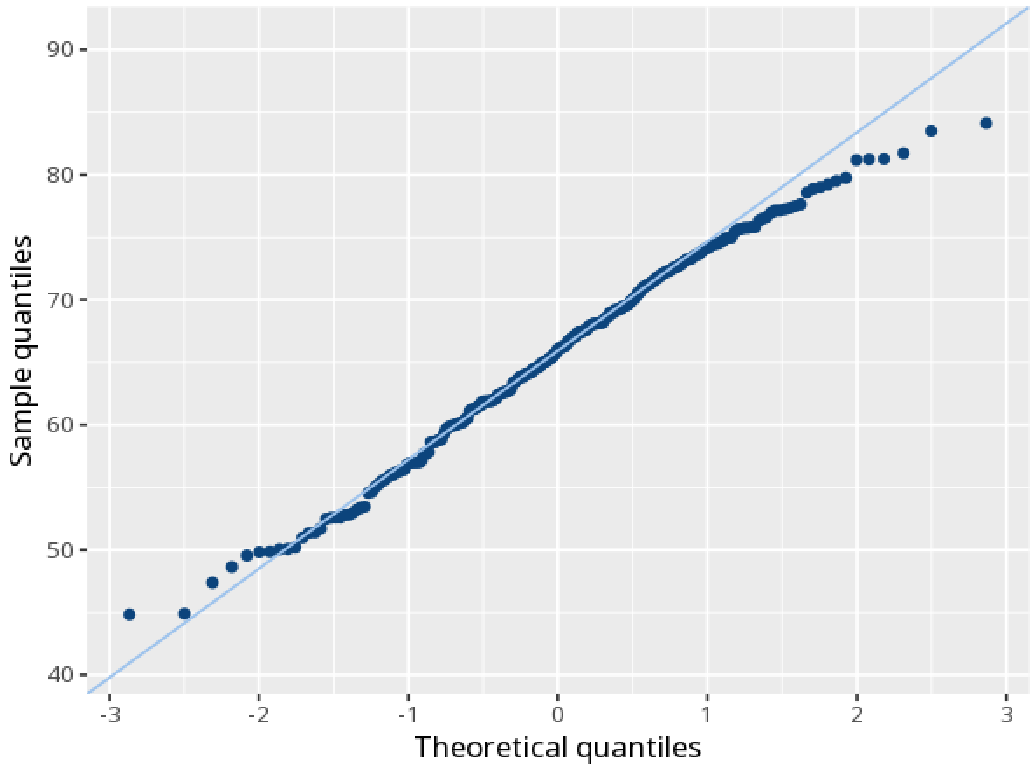
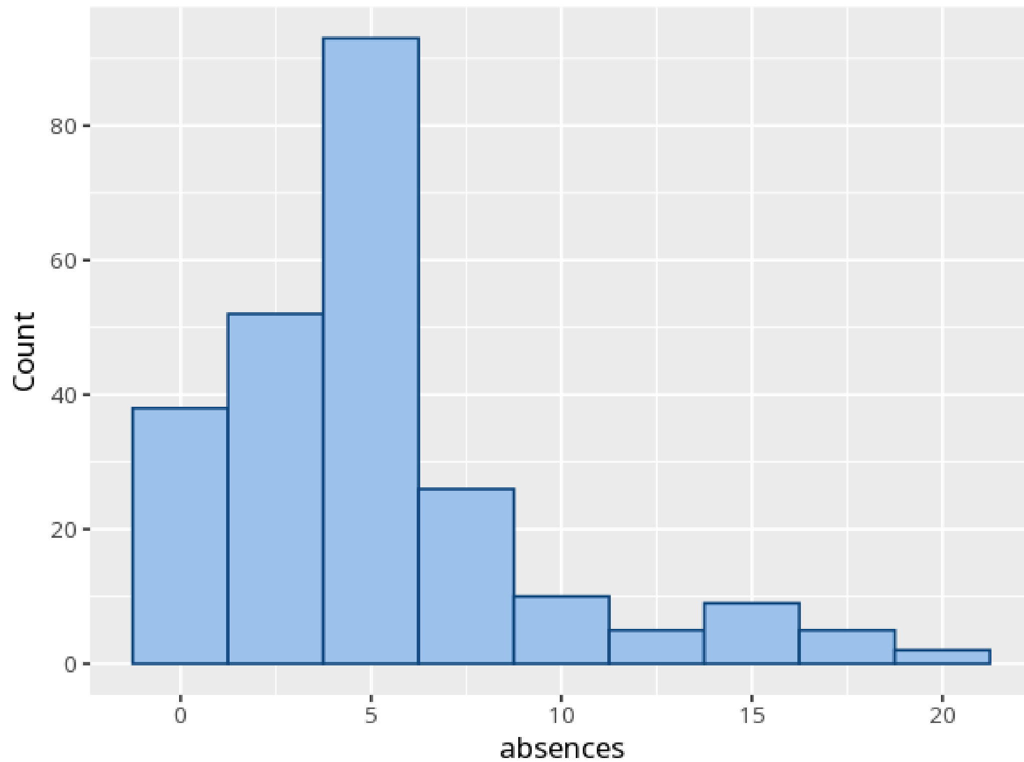
1 Distribution & normality

Variable	Test	Statistic	N	p	Skew	Kurtosis (excess)
exam_score	Shapiro-Wilk	W 0.99	240	.158	−0.18	−0.53
exam_score	K–S (Lilliefors)	D 0.04	240	.608	−0.18	−0.53
absences	Shapiro-Wilk	W 0.86	240	<.001	1.50	2.43
absences	K–S (Lilliefors)	D 0.17	240	<.001	1.50	2.43

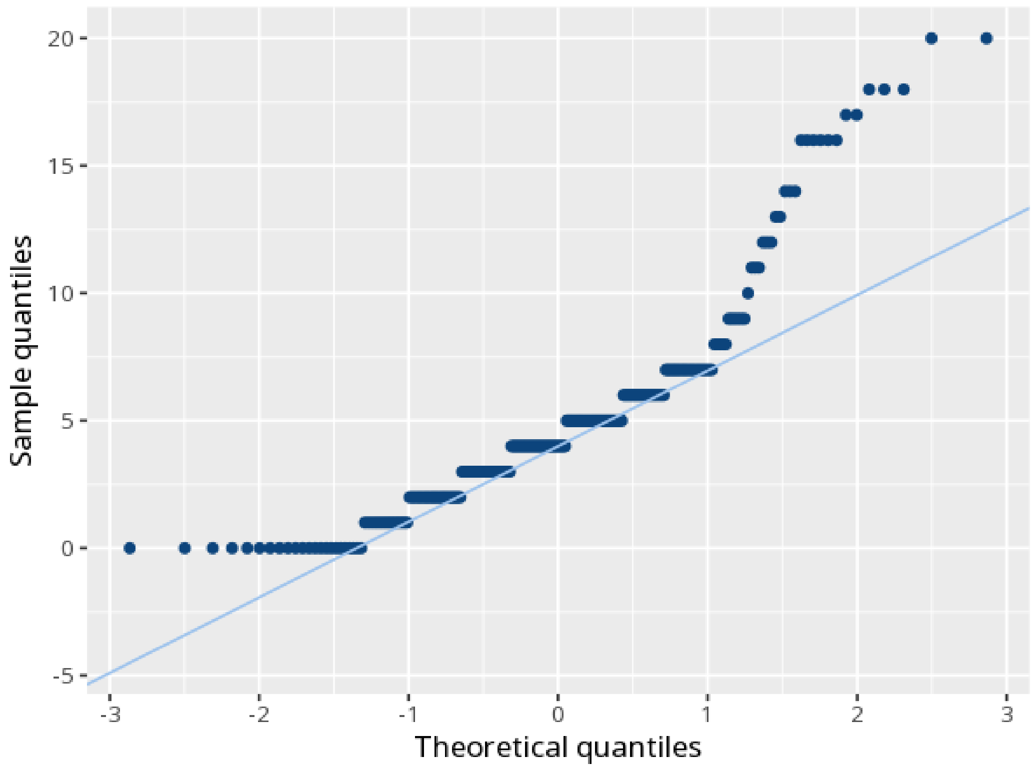




90
80
70
60
50
40
Sample quantiles
-3
-2
-1
0
1
2
3
Theoretical quantiles



80
60
40
20
0
Count
0
5
10
15
20
absences



A small p (below .05) suggests the data depart from normality. With large samples these tests over-flag trivial deviations; with small samples they have low power, so a p > .05 does not prove normality - it only means no departure was detected. Either way, judge normality mainly from the Q–Q plot - points hugging the diagonal indicate normality.

exam_score: Normality was assessed with the Shapiro-Wilk test, W = .99, p = .158. absences: Normality was assessed with the Shapiro-Wilk test, W = .86, p < .001.

Statistical basis: Shapiro-Wilk test for normality (3-5000 cases) (Shapiro, S. S., Wilk, M. B. (1965). "An analysis of variance test for normality (complete samples)." Biometrika, 52(3/4), 591-611.); Additional normality tests (Lilliefors and related) (Gross J, Ligges U (2015). nortest: Tests for Normality. R package.).